An Adaptive Self-Healing Framework for Quantum Machine Learning

Abstract

Variational Quantum Circuits (VQCs), the trainable core of most near-term Quantum Machine Learning (QML) models, are notoriously fragile to train, with barren plateaus, poorly chosen learning rates, and unstable optimizer settings routinely derailing training runs. Existing responses are either manual and expert-dependent, relying on hand-tuned hyperparameter schedules, or automated but non-adaptive, such as one-shot AutoML search or fixed rule-based repair; none treat repair selection itself as a decision problem that improves from accumulated outcomes. This paper proposes an Adaptive Self-Healing Framework for QML that closes a continuous monitor-diagnose-repair-verify loop around VQC training. A health-monitoring system extracts training telemetry, a rule-based detector coupled with a confidence-gated diagnosis engine identifies the likely failure mode, and a six-arm repair library encodes candidate fixes as pure configuration transforms. As the central contribution, repair selection is formalized as a contextual Thompson Sampling bandit with per-arm Bayesian linear regression posteriors, updated online from a multi-objective reward signal, with a digital-twin proxy trainer vetting each repair before deployment. A complete vertical slice, implemented in PennyLane, has been validated end-to-end across all fourteen architectural phases for a representative failure path, ahead of a planned ablation study comparing random, greedy, rule-based, and adaptive policies to quantify gains in recovery sample efficiency.

Literature Survey

S.no	Author(s) / Year	Title / Focus	Methodology	Key Findings	Research Gap
1	McClean et al. (2018)	Barren plateaus in quantum neural network training landscapes	Theoretical analysis of gradient variance in parameterized quantum circuits	Gradient variance vanishes exponentially with qubit count for 2-design circuits, causing untrainability	Purely diagnostic; no runtime detection or corrective mechanism proposed
2	Cerezo et al. (2021)	Cost function dependent barren plateaus in shallow parametrized quantum circuits	Analytical study of cost-function locality effects on trainability	Even shallow circuits suffer barren plateaus if a global cost function is used; local cost functions delay the effect	Prescribes a static design choice, not an adaptive runtime response to detected failures
3	Cunningham & Fitzsimons (2024)	Investigating and Mitigating Barren Plateaus in VQCs: A Survey	Systematic literature review and taxonomy construction	Organizes existing BP mitigations into five categories (initialization, architecture, cost function, training strategy, other)	Mitigation is treated as a static, human-selected choice with no mechanism to choose among competing fixes automatically
4	Ganek & Corbi (2003)	The Dawning of the Autonomic Computing Era	Introduced the MAPE-K control-loop architecture	Established Monitor–Analyze–Plan–Execute–	Domain-general; not concerned with statistical/ML

				Knowledge as the reference model for self-managing systems	training pathologies
5	Gorla (IMDEA survey)	Achieving Cost-Effective Software Reliability Through Self-Healing	Survey of self-healing software techniques	Identifies fault localization and healing-strategy selection as the two hardest sub-problems in MAPE-K	Notes human expertise is hard to replace in diagnosis, a gap unaddressed by rule-only systems
6	Zhang, Zhai, Ma & Shen (2021)	AUTOTRAINER: An Automatic DNN Training Problem Detection and Repair System	Periodic monitoring of loss/accuracy/gradient telemetry against a rule-based catalogue of 5 known problems, with a fixed built-in fix per problem	100% detection rate, no false positives; fixed 97.33% of problem instances, +47.08% accuracy on average	Repair choice per diagnosed problem is deterministic and fixed — never adapts based on which fix worked best historically
7	Wardat, Cruz, Le & Rajan (2022)	DeepDiagnosis: Automatically Diagnosing Faults and Recommending Actionable Fixes in DL Programs	Fault diagnosis engine separated from hyperparameter-optimization search	Treats diagnosis (why a model fails) as distinct from search (finding a better config from scratch)	Diagnosis and repair remain separate, non-learning stages with no outcome memory
8	Agrawal & Goyal (2013)	Thompson Sampling for Contextual Bandits with Linear Payoffs	First rigorous regret analysis of Thompson Sampling under linear-payoff, adaptive-adversary contexts	Proved a regret bound of $\tilde{O}(d^{1.5}\sqrt{T})$, giving contextual TS formal theoretical grounding	Assumes linear reward structure; not previously applied to a repair-selection or QML setting
9	Chapelle & Li (2011)	An Empirical Evaluation of Thompson Sampling	Empirical comparison of TS against UCB-style bandit algorithms	TS empirically outperforms UCB in practice despite historically weaker theoretical bounds	Domain-general evaluation; never tested in a training-repair or quantum-computing context
10	Tao, Xu & You (2025)	Provable Performance Bounds for Digital Twin-Driven DRL in Wireless Networks	Bisimulation-metric-based bound on digital-twin/real-world discrepancy	Provides theoretical justification for trusting twin-based pre-evaluation before real deployment	Bounds are network-specific; no equivalent safety bound exists yet for a QML training-repair proxy twin

Objectives

- To design and implement a health-monitoring and diagnosis subsystem for Quantum Machine Learning training — comprising a lightweight telemetry-profiling layer (loss trajectory, gradient statistics, parameter norms, optimizer state) for a PennyLane VQC training loop, a rule-based failure detector, a probabilistic diagnosis engine with confidence gating, and a cosine-similarity-searchable failure-embedding memory — enabling accurate, low-overhead identification of training pathologies such as barren plateaus and learning-rate misconfiguration.
- To formalize a repair library and a safe pre-deployment verification pipeline, consisting of at least six candidate repair actions expressed as pure, side-effect-free configuration transforms (e.g., learning-rate reduction, optimizer substitution, gradient clipping, re-initialization, ansatz simplification, batch-size adjustment), combined with an oscillation/safety filter and a short-horizon digital-twin proxy trainer that vet each candidate repair before it is committed to the live training run.
- To formalize repair-strategy selection as a contextual Thompson Sampling bandit problem, implementing per-arm Bayesian linear regression posteriors over a shared context vector derived from diagnosis output and training telemetry, a multi-objective reward fusion function combining loss improvement, gradient-health recovery, training-time cost, and stability, and dual online learning through live posterior updates and persistent knowledge-base storage — positioning this adaptive bandit formulation as the central novelty distinguishing the framework from static, rule-based repair.
- To implement safe, reversible deployment of repairs with quantitative learning evaluation, including rollback of repairs that underperform their digital-twin prediction and instrumentation of cumulative regret relative to an oracle policy, so that the efficiency and reliability of the adaptive repair policy can be measured objectively over time.
- To empirically validate the complete framework and document it to academic research standards, by demonstrating all fourteen architectural phases end-to-end on a concrete VQC classification task for at least one full failure path, conducting a statistically meaningful ablation study (following QNode vectorization) comparing random, greedy, rule-based, and adaptive repair policies, and producing a formal Software Requirements Specification and supporting documentation suitable for thesis or publication-track review.